RECOMMENDED OUTLINE

Report of Compliance with the Policy on Development Data Quality

[*Name of the product/process/system being reported on*]

PRODUCTION AND SUBMISSION	
BUSINESS UNIT(S):	*Identification of the business unit(s) in charge of the data product, process, system or tool being reported on.*
AUTHORS:	*Name and title of authors of the report*
CONTRIBUTORS:	*Name, title, and role of other contributors*
ENDORSED BY:	*Name and affiliation of the Director(s) or Manager(s) endorsing the report*
ENDORSED ON:	*Date the report was endorsed*
VERSION:	*Version of the report. It is recommended to use semantic versioning (e.g., 1.0.0)*
NOTE:	*Any additional critical information on the production or submission*
REVIEW BY THE OFFICE OF THE CHIEF STATISTICIAN	
REVIEWED ON:	*Date the report was reviewed by the Office of the Chief Statistician (OCS)*
REVIEWED BY:	*Name and title of the member(s) of the OCS who reviewed the report*
OUTCOME:	*Summary of key issues and conclusions of the review by the OCS. More detailed information will be provided in the report (see section 4 of the recommended outline).*

Data Quality Compliance Report

Report

DRAFT – 2025-10-30

Guidance for Producing a Report of Compliance with the Policy on Development Data Quality

Report in HTML

A searchable version of the report.

Report in PDF

A downloadable version of the report.

Source Code

View the source code on GitHub.

Acknowledgments

The document was written by Olivier Dupriez, Deputy Chief Statistician, Development Data Group (DECDG) and Office of the World Bank Group Chief Statistician. It benefited from comments and suggestions by [...add list after review process]

Use of Artificial Intelligence (AI)

AI tools, including ChatGPT 4 by OpenAI and Microsoft Copilot's Auto Rewrite tool, were utilized in the development of this document. The initial content was drafted by the author; AI was subsequently used to enhance or edit the text for clarity and consistency. All AI-generated content was thoroughly reviewed for accuracy and relevance.

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Maintenance

> This Guide is maintained on GitHub as a markdown document, allowing for version control, collaborative editing, and automatic conversion to HTML and PDF formats. GitHub repository: https://github.com/worldbank/data-quality-compliance-report

Preface

In 2024, the World Bank Group ("the Bank") adopted the Development Data Quality Policy to enhance the quality, transparency, and usability of the development data it produces, curates, uses, and disseminates. The Policy applies to all Bank development data and establishes clear expectations for robust quality control, assurance, and continuous improvement across the entire data lifecycle.

The Bank's data ecosystem is intentionally decentralized. Data collection, acquisition, production, curation, and dissemination are carried out by specialized business units across the institution, enabling flexibility and responsiveness to diverse operational and analytical needs. However, decentralization also creates challenges for maintaining consistency, interoperability, and adherence to common standards. The Policy addresses these challenges by promoting alignment with recognized statistical, data management, and metadata standards and best practices.

The World Bank Group Chief Statistician coordinates the development of standards, technical guidance, and reference materials to support units in designing and managing data activities in line with policy requirements. Ultimately, though, **responsibility for implementing the Development Data Quality Policy rests with the business units** that create and manage development data assets.

While independent evaluations or audits may be used to support policy oversight, **implementation primarily relies on a structured process of self-assessment and reporting**. Business units are expected to prepare a *Report on Compliance with the Policy on Development Data Quality* for each core data program, product, processes, or systems under their accountability. These reports summarize self-assessment results and demonstrate the unit's commitment to data quality and continuous improvement.

This guide explains the purpose and production of these reports, outlines a suggested structure, and lists what information each section should contain. Specific guidance is provided separately for the assessment and reporting on the quality of modeled estimates. See the Bank's *Quality Assurance Framework for Modeled Estimates of Development Data*.[1]

Purpose of the Compliance Reports

Compliance reports serve two key purposes, addressing both internal and external needs:

- **For the business unit and the Bank**: The reports provide an opportunity to reflect on current practices, document adherence to quality standards, and identify areas for improvement. They serve as a tool for internal learning, continuous improvement, and informed decision-making regarding data processes and resource needs.

- **For data users**: The reports demonstrate transparency in how data is managed and maintained, helping to build trust in the quality and integrity of the Bank's development data. They signal a clear commitment to accountability and good data governance in the Bank's development data activities.

Initiation of Compliance Reports

The production or update of compliance reports is typically initiated by the business units themselves, as part of their ongoing commitment to data quality and transparency. However, in some cases, the preparation of a report may be prompted by a specific request from Bank leadership, the Data Governance Council, or the Chief Statistician.

Compliance reports may also be initiated in response to special circumstances, such as when the quality or integrity of Bank data is questioned by external stakeholders, or when a major new data partnership is considered.

Scope of the Compliance Reports

Compliance reports focus on how specific development data activities align with the requirements and principles set out in the Development Data Quality Policy. The exact scope of compliance reports is determined by the business unit preparing the report, in consultation with the requestor when applicable.

In practice, compliance reports will focus on one or several of the following:

- **Data products** (e.g., specific datasets)
- **Processes** (the processes used to generate and manage development data at any step of the data life cycle, e.g., data collection, validation, documentation, or dissemination workflows)
- **Systems and tools** (e.g., data dissemination platforms, processing tools, or data management systems)

Institutional arrangements related to development data activities (such as Bank-wide governance structures, staff responsibilities, or administrative directives) may be within the scope of the Development Data Quality Policy but will not be the object of self-compliance reports. They should however be referenced when relevant to understanding the context or explaining how quality is managed for specific data products, processes, systems and tools.

Scope in Relation to Third-Party Data

Compliance reports cover only development data for which the Bank holds responsibility—i.e., data it produces, transforms, curates, maintains, or disseminates. Data obtained from third parties falls outside the direct scope of the Development Data Quality Policy in terms of assessing their intrinsic quality. However, the decision to use third-party data, or to redistribute them, and the whole process of transformation of third-party data for the production of Bank data, is within the scope of the Policy and must be guided by appropriate due diligence.

Business units are expected to assess whether third-party data are fit for their intended purpose and to document the process by which this determination was made. Compliance reports should not evaluate the quality of third-party data per se, but must describe the procedures and criteria used to review, select, and, where applicable, validate or annotate such data before use or redistribution. This includes documenting any efforts to understand the source, methodology, and limitations of the third-party data, in line with the Bank's responsibility to ensure transparency and accountability in its data-related decisions.

Who Is Responsible for Producing Compliance Reports?

The Development Data Quality Policy is implemented mainly through decentralized accountability and relies primarily on self-assessment by business units. Accordingly, compliance reports are not audits or independent evaluations—they are self-reported assessments produced by those most familiar with the data product, process, or system being reviewed.

Each compliance report is authored by staff within the responsible business unit(s), ideally by those directly involved in or closely overseeing the relevant work. Authors are expected to conduct an honest, objective, and thorough assessment. While input from other Bank colleagues (including from the Chief Statistician's Office) or external stakeholders may be sought to inform the assessment, the report must be prepared and owned by the business unit(s) in charge.

Compliance reports must be formally endorsed by a manager or director within the responsible business unit(s).

Use of the Recommended Outline

Where possible, the recommended outline provided in this document will be used to structure compliance reports. While strictly following this outline is not mandatory, it is strongly encouraged to ensure consistency across reports.

Within each section and sub-section, concise descriptions should be provided and references or links to relevant documents, websites, or other sources that support the information presented should be provided.

The proposed outline presented in section 2 includes, for each of the thirteen core principles of data development quality (sections 2.1 to 2.13), a set of suggested items to help authors identify topics that may be addressed. These lists are only intended as guidance; not all items will be applicable in every situation.

The italicized reminders of each principle in the report should be retained in compliance reports, as these help readers who are not familiar with the Development Data Quality Policy to understand the context and purpose of the report.

Length of the Compliance Report

The length of a compliance report will vary depending on the availability of existing documentation. Business units that already have thorough, openly accessible documentation of their processes and products can produce shorter reports by referencing these resources instead of describing procedures in detail. Where such documentation does not exist, the compliance report becomes an opportunity to create it. In such cases, units may choose either to produce a longer report that includes the necessary detail or to prepare technical documentation separately and have the report reference it.

References and links alone are not sufficient; reports must include enough explanation to be self-contained and understandable without requiring readers to consult external sources.

Role of the Office of the Chief Statistician

To support consistency and quality, the final version of the reports must be reviewed by the Office of the Chief Statistician. The review by the Office of the Chief Statistician's does not constitute formal endorsement of the Report.

The reports can be edited after review by the Office of the Chief Statistician, whose role is mainly to support business units in their production of informative and credible reports on quality that will enhance the credibility of their data products, processes, and systems and tools, and to help them identify areas for improvement.

The Chief Statistician may object to the publishing of a Compliance Report if it is considered incomplete, inaccurate, or potentially misleading.

When Should Compliance Reports Be Produced or Updated?

Compliance reports are typically produced after a major stage in the data lifecycle, such as the implementation of a new data process, the release (or near-release) of a new (version of a) data product, or the deployment of a new system or tool. While reports are usually generated soon after completion or launch, they may also be produced retrospectively for existing or legacy data products, processes, or systems when a review is needed or requested. They serve as a reflection on what was done and why, how quality was managed, and how well the activity aligns with the Development Data Quality Policy.

Compliance reports are not intended for use during the design phase. However, the self-assessment criteria and guidance provided in this note may serve as a valuable reference to informing the design of future data activities and ensure they are aligned with quality standards from the outset.

How Frequently Should Compliance Reports Be Updated?

For ongoing or recurring data products, processes, or systems, an initial compliance report should be prepared to document the current state as of the first review—whether at launch or for an already existing version. Subsequent updates to the report should be produced whenever significant changes are made to the product, process, or system. An update may also be triggered by a formal request from management, the Chief Statistician, or other stakeholders.

When an updated compliance report is produced, it must clearly document what has changed since the previous version. To ensure transparency, the report's introduction must then include a dedicated sub-section summarizing all major changes and additions compared to earlier reports.

How Will Compliance Reports Be Shared and Stored?

Compliance reports are intended to be made publicly available when the associated data product, process, or system is publicly accessible. If the subject of the report is not public, the report will be designated as internal and marked for official use only.

Public reports will be published online in a dedicated section on Data Quality of a website maintained by the Office of the Chief Statistician. In addition, business units may choose to share the reports through other relevant channels when appropriate.

The Office of the Chief Statistician will maintain a complete archive of all compliance reports—both public and internal—to ensure institutional memory and enable future reference.

Where to Find Guidance on Good Practices and Compliance?

To support compliance with the Development Data Quality Policy, a range of technical resources is available which describe good practices and standards adopted by the Bank. These include:

- The *Implementation Guide to the Development Data Quality Policy*. This document provides references to multiple more specialized technical recommendations. [DRAFT]
- The *Quality Assurance Framework for Modeled Estimates of Development Data.* [DRAFT]
- The *Reference Guide on Metadata Standards for Development Data*. [DRAFT]
- The *Guide on Metadata Quality Assessment and Enhancement for Development Data*. [FORTHCOMING]
- The *Guide on Anomaly Detection and Data Augmentation for Time Series*. [FORTHCOMING]
- The Development Data Quality website, maintained by the Office of the Chief Statistician. [FORTHCOMING]

Business units are encouraged to consult these resources when preparing their reports or designing data-related activities. For additional clarification or tailored support, the Office of the Chief Statistician is available for consultation.

1. Introduction

If a report does not include Section 1.2 (Update Rationale), Section 1.1 (Background) should not be created as a standalone subsection. In such cases, the background information will constitute the Introduction in full, without further subdivision

1.1 Background

The Background section of the Introduction should provide information describing the product, process, or system/tool being assessed. It should clearly but briefly answer the following key questions:

- What is being reported on? (identification of the product, process, system or tool)
- Who is responsible for it? (identification of the business unit(s) and key partners if any)
- Why was it created or implemented? (objective of the product, process, system or tool)
- For whom is it intended? (intended primary stakeholders of the product, process, system or tool)
- Since when has it been in place or in use? (origin of the product, process, system or tool)
- Who asked for the report? (self-decision by the responsible business unit, specific request, or required in response to criticism)

1.2 Update Rationale

If the report updates a previous version, include this subsection titled "Update rationale". It should explain the rationale for producing a new version and highlight the main changes or additions compared to the earlier report. Where appropriate, this can be presented in the form of a table that serves as a change log.

2. Compliance with the Core Data Quality Principles

The Bank's Development Data Quality Policy defines thirteen core principles that guide the quality of data products, processes, systems and tools, as well as the institutional arrangements that support them. Compliance with each principle will be self-reported by the business unit in charge in the thirteen sub-sections below.

For each principle, a list of topics and questions that may be considered for inclusion is proposed in the sections below. Not all will apply in every circumstance, as some pertain only to data products, processes, or systems and tools. When an item is applicable, a description of the self-assessed status of compliance (indicating clearly full, partial, or non-compliance) should be provided, along with supporting evidence. Provide relevant links to materials like documents or websites.

2.1 Impartiality and Independence

2.1.1 Principle overview

"Bank Development Data shall be produced professionally, impartially and free from inappropriate influence." (Principle 1.a of the Policy)

Data must be produced according to the highest professional standards, ensuring that methods, sources, and analytical choices are grounded in sound science and best practices. The data must be impartial, reflecting objective reality rather than political, institutional, or personal interests. To maintain trust and credibility, the data production process must be protected from inappropriate influence at all stages, including design, collection, processing, analysis, and dissemination.

2.1.2 Instructions

This principle applies to **data products** and **data processes**.

This section of the report should explain how:

- professional and methodological standards are upheld,
- outputs reflect objective reality without bias, and
- inappropriate influence is prevented and managed.

Authors must describe both institutional safeguards that protect impartiality, and unit-specific practices and operational controls.

Use clear evidence such as documents, workflows, governance mandates, reviews, versioned methodologies, training records, or examples from recent production cycles.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Institutional Environment, Oversight, and Safeguards**

Explain how the Bank-wide environment supports impartial, high-quality data. Include:

- Policies safeguarding independence of statistical/analytical work.
- Organizational separation between data production and policy/advocacy.
- Formal commitments to ethics and professional standards.
- Conflict-of-interest policies and prevention and mitigation procedures.
- Reporting mechanisms for undue influence, pressure, or manipulation attempts.

- Internal or external audits reviewing impartiality safeguards.
- Staff training and onboarding on ethics, quality, and independence.

Clearly identify the sources of funding for the data product, process, or system/tool. If any external funding sources could potentially present concerns, outline the procedures established to ensure the business unit's professional independence.

If relevant, include the following paragraph or an adaptation of it, and indicate how staff, consultants, and partners are reminded of these obligations in the context of the product, process, or system being reported on. This will mainly apply of data collection and dissemination programs that generate sensitive or highly impactful datasets.

> *The World Bank Group protects the integrity and independence of its data activities through policies, directives, and procedures that include:*

- ***Ethics and training*** *– Mandatory onboarding and refresher training reinforce professional standards.*
- ***Conflict of interest controls*** *– Institutional safeguards and contractual clauses prevent political, financial, or institutional interference. Paragraph V, section 5 (c) of the Articles of Agreement of the IBRD states that "The President, officers and staff of the Bank, in the discharge of their offices, owe their duty entirely to the Bank and to no other authority. Each member of the Bank shall respect the international character of this duty and shall refrain from all attempts to influence any of them in the discharge of their duties."*
- ***Whistleblower protections*** *– Secure, anonymous reporting channels are managed by the Ethics and Business Conduct (EBC) Department. Staff members have several independent internal mechanisms at their disposal to report undue pressure from other staff members, including the Anti-Harassment Coordinator and the EBC Investigation Unit. EBC has no jurisdiction over the Board (Executive Directors, Alternate Directors, Senior Advisers, and Advisers). However, staff members can report matters to the Ethics Committee of the Board, directly or via EBC. EBC and the Ethics Committee procedures protect anonymity when staff report concerns.*

2. **Data Design**

Describe how design decisions ensure validity and impartiality:

- Show that methods and indicators are selected based on scientific rigor and fitness for purpose.
- Reference internationally recognized standards and best practices where applicable.
- Explain how methodological changes are decided and demonstrate freedom from external pressure.
- Provide evidence of peer/expert review before implementation.

3. **Data Collection**

Demonstrate independence and methodological rigor in collection:

- Provide evidence of representative, accurate sampling (as relevant0 and collection methods.
- Describe safeguards against stakeholder/interest-group influence.
- Document impartiality selection of, and training for data collectors.
- Describe independent quality monitoring during collection.

4. **Data Processing**

Ensure objectivity and reproducibility:

- Confirm that no data is altered or excluded without documented, valid reasons.

- Mention use of automation and guarantees of reproducibility to reduce subjective decisions where appropriate.
- Describe independent validation checks.

5. **Data Analysis**

Confirm analytical rigor and neutrality:

- Show that analytical approaches are chosen for relevance and methodological strength.
- Confirm that no selective use of data/methods is employed to obtain preferred results.
- Indicate whether multiple perspectives are considered in interpretation.
- Document peer review of analysis for objectivity.

6. **Data Dissemination**

Protect impartiality of communication and access:

- Describe procedures ensuring release according to pre-announced schedules, without political delay.
- Confirm equal access to the data product for all users; if embargoed, explain the reason for embargoes and confirm that no early access is given to parties who could influence interpretation.
- Indicate whether outputs include methodological notes, limitations, and uncertainty statements.
- Document the sign-off process for final outputs.

7. **Partnerships and Governance**

If applicable:

- Explain external partner selection, vetting, and governance arrangements (if any; otherwise mention that the product/process is entirely designed and implemented internally).

8. **Addressing Risks and Incidents**

If applicable:

- Summarize incidents or risks related to impartiality/independence, describing how they were mitigated. Do not name individuals; focus on describing the issue and resolution.

2.2 Responsible Data Management

2.2.1 Principle overview

"The Bank shall ensure that its activities that are part of the Data Management Life Cycle are designed and conducted responsibly, taking into account potential ethical, legal and information technology risks." (Principle 1.b of the Policy)

The Bank is committed to managing data responsibly throughout the entire data life cycle—from acquisition and storage to processing, use, and disposal. This includes proactively identifying and addressing potential ethical, legal, and technological risks associated with data handling. Responsible data management requires implementing safeguards to protect privacy, ensure security, and uphold legal and ethical standards.

2.2.2 Instructions

This principle applies to **data products**, **data processes**, and **systems/tools**.

The compliance report should clearly demonstrate:

- how risks related to privacy, security, fairness, and legality were identified and mitigated,
- how ethical and legal safeguards were implemented throughout the lifecycle,
- how accountability and responsible data stewardship were ensured, and
- how compliance with relevant Bank policies and standards was evidenced.

Provide concrete examples, evidence, and references to policies, procedures, security controls, audit results, access logs, and risk assessments.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Data Acquisition**

Confirm acquisition practices ensure legality, authorization, transparency of source, and ethical handling of data:

- Confirm that all data sources used as input are lawful and authorized.
- State whether acquisition followed the *Procedure on Data Acquisition, Archiving, and Dissemination* or the *Procedure on Data Procurement*.
- Confirm that all datasets (inputs and outputs) are classified by confidentiality, licensing, and copyright status, and that this classification is documented and up to date.
- Ensure every dataset's source is fully traceable (mode of acquisition, source organization, acquisition date, and any pre-use processing).

- Describe procedures for validating licensing terms and usage rights for all third-party datasets, including documentation of permitted uses and restrictions.
- Provide proof of consent for personal or sensitive data where required.
- State whether an ethical implications review (fairness, potential harm) was conducted before data collection or acquisition.

2. **Data Storage**

Ensure data is stored securely with access controls appropriate to sensitivity and legal/ethical obligations:

- Confirm that secure storage systems are used with appropriate access controls.
- Indicate whether sensitive or personal data are encrypted at rest.
- If personal information is handled, confirm consultation with the Data Privacy Office (DPO) and full implementation of its recommendations.
- Describe user access permissions and roles, and how they are reviewed and kept current.
- Confirm that all data versions are archived with version-specific metadata (dates, change descriptions).
- Describe monitoring and logging procedures for access to stored data.

3. **Data Processing and Use**

Ensure data is processed solely for authorized purposes with safeguards against misuse, inaccuracy, or harm:

- Confirm that processing activities are documented and match their intended purposes.
- Describe measures to prevent unauthorized processing or misuse.
- Identify potential fairness or bias issues in processing and how they are mitigated.
- Outline technical safeguards ensuring data integrity during processing.
- Confirm that data use is limited to defined, authorized purposes.
- Specify access and licensing conditions for output datasets, and how these are communicated to users.
- Document data sharing agreements and applicable terms of use.
- Identify ethical concerns in data applications and describe mitigation measures.
- Explain privacy protection measures during analysis and dissemination.

4. **Data Retention and Disposal**

Ensure retention and disposal practices meet policy requirements, prevent unauthorized exposure, and respect rights:

- Confirm existence of a documented data retention schedule.
- Describe secure deletion or anonymization procedures after the retention period.
- State whether backups and copies are also securely disposed of.
- Provide or reference the procedure ensuring disposal meets legal and contractual obligations.

5. **Risk Management and Safeguards**

Demonstrate risks are continuously monitored and mitigated through appropriate controls and preparedness measures:

- Identify ethical, legal, and technological risks, and describe mitigation measures (technical, organizational, procedural).
- Reference the incident response plan for data breaches or misuse.
- State whether regular security assessments and vulnerability testing are conducted.

6. **Governance and Accountability**

Show that responsibilities, oversight, and reporting channels are clearly defined to enforce compliance:

- Identify roles and responsibilities for data management.
- Confirm that data management policies and guidelines are current and available.
- Describe training provided to staff on responsible data handling.
- Indicate mechanisms for reporting and addressing non-compliance or incidents.
- Mention periodic internal reviews or audits of data management practices.

2.3 Verifiability

2.3.1 Principle overview

"Bank Development Data shall be verifiable. The Bank shall ensure that, among other things, its Data Processes are designed and documented accordingly." (Principle 1.c of the Policy)

Bank development data must be verifiable, meaning that others can trace, understand, and assess how the data were produced. To support this, the Bank ensures that its data processes are designed with transparency in mind and are thoroughly documented at every stage—from data collection and processing to analysis and dissemination. Clear documentation of methodologies, sources, and transformations enables reproducibility, facilitates quality assurance, and strengthens user confidence.

2.3.2 Instructions

This principle applies to **data products** and **data processes**.

This section should demonstrate how:

- the data production process is fully traceable,
- methods, sources, and transformations are clearly documented, and
- reproducibility is technically and operationally supported.

Include references to publicly available documentation when applicable, and provide internal evidence where access is restricted (e.g., ticket systems, code repositories, standard operating procedures).

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Methodology Documentation**

Show that all methodological components can be understood and evaluated:

- Provide references to complete and up-to-date methodological documentation.
- Include links or attachments to definitions, concepts, classifications, and standards applied.
- Explain and justify methodological choices made.
- Provide references to the documentation of the data collection design (e.g., sampling methods, questionnaires, instruments).
- Report limitations, assumptions, and uncertainties, with supporting evidence.

2. **Data Sources**

Demonstrate full transparency and traceability of inputs:

- Submit a current list of primary and secondary data sources.
- Provide documentation of quality assessments conducted on these sources.
- If applicable, include evidence of version histories for all source datasets.

3. **Data Collection**

Ensure users can verify how raw data were obtained:

- Demonstrate that original, unaltered data are preserved.
- Provide records of collection procedures, tools, and timelines.
- Document any modifications made during collection (e.g., replacements, corrections).
- Confirm the availability of metadata describing collection conditions and constraints.
- Show evidence of a documented chain of custody for collected data.

4. **Data Processing and Transformation**

Provide a transparent, reproducible transformation path:

- Describe and document processing workflows step by step.
- Provide records of, or references to, all data cleaning, coding, and transformation rules.
- Document the algorithms, formulas, or statistical models applied.
- Submit evidence of version control for code/scripts used in processing.
- Confirm the availability of logs showing all changes from raw data to final dataset.

5. **Data Analysis**

Make analytical decisions and outputs fully assessable:

- Describe the analytical methods and tools applied, with references.
- Provide detailed explanations of indicator calculations and derived variables.
- Confirm that intermediate analysis files and scripts are preserved, and describe how and where they are stored.
- Document quality checks and validation procedures applied.
- Provide evidence of reproducibility checks by independent staff or reviewers.
- Confirm that any reproducibility package is current, complete, and functional.

6. **Data Dissemination**

Ensure users can verify what was released and how it was constructed:

- Provide references to the methodological notes and metadata made available with the datasets.
- Describe the file formats, data structures, and variable definitions used.
- Confirm use of persistent identifiers for datasets and dataset versions.

- Provide evidence that all released versions of datasets have been archived with change logs.

2.4 Transparency

2.4.1 Principle overview

"The Bank shall openly and promptly communicate on any identified material quality issues with respect to Bank Development Data." (Principle 1.d of the Policy)

The Bank upholds transparency by openly and promptly communicating any material quality issues identified in its development data. This includes disclosing limitations, errors, revisions, or uncertainties that may affect the interpretation or reliability of the data.

2.4.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

This section should clearly show how:

- material quality issues are detected and documented,
- limitations, uncertainty, and risks are communicated proactively,
- users are informed promptly of errors or revisions, and
- accountability and continuous improvement are supported.

Provide concrete examples, evidence, and references to policies, workflows, publication pages, and versioned documentation.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Identification of Quality Issues**

Demonstrate mechanisms exist to detect and record significant issues:

- State the criteria used to define what constitutes a "material" issue.
- Describe procedures in place to detect and document material quality issues.
- Explain the process for escalating quality concerns internally.
- Confirm the existence of logs or records of identified limitations, errors, or uncertainties.
- Indicate whether regular reviews of data quality indicators are conducted to flag emerging issues.

2. **Documentation of Limitations and Uncertainty**

Ensure users can understand potential impacts on interpretation:

- Provide clear explanations of known limitations in scope, coverage, timeliness, or accuracy.
- State whether quantitative measures of uncertainty (e.g., confidence intervals, margins of error) are produced and made publicly available where applicable.
- Document potential sources of bias or error.
- Explain how limitations are presented to users, including potential impacts on interpretation and use.
- Confirm that methodological notes describing quality issues in context are accessible to all data users.

3. **Communication of Errors and Revisions**

Show how corrections are handled with openness and timeliness:

- Confirm whether a revision policy is publicly available, describing procedures and triggers for revisions.
- Describe how users are promptly informed when errors are discovered and corrected.
- State whether an errata system or equivalent mechanism exists to report and correct errors.
- Confirm that revised datasets and products are clearly labeled with dates and version numbers, and whether semantic versioning is applied.
- Provide evidence that change logs are available documenting what was corrected and why.
- Indicate whether a public archive of previous versions is maintained.
- If relevant, describe past cases where significant quality issues were communicated, including how this was done.

4. **User-Facing Transparency Mechanisms**

Ensure users can find and understand quality information:

- Confirm that quality notes and disclaimers are included in published datasets, reports, or dashboards.
- Indicate whether limitations are described in plain language for non-technical audiences.
- Describe channels for users to submit questions or report suspected issues.
- Outline the process for technical review and communication if estimates are challenged by stakeholders or external parties, and how outcomes are shared.
- Confirm whether transparent response procedures exist for user feedback.

5. **Internal Processes for Transparency**

Demonstrate organizational readiness and accountability:

- Define roles and responsibilities for reviewing and approving quality disclosures.
- State whether staff are trained to communicate data limitations clearly and accurately.
- Confirm that transparency requirements are integrated into project workflows.

6. **Accountability and Continuous Improvement**

Support institutional learning and trust:

- Indicate whether there is a public statement of commitment to transparency in data quality for the product or process, and provide a link.

> Describe any mechanisms for independent review of transparency practices.

2.5 Relevance

2.5.1 Principle overview

"The Bank aims to achieve that all Bank Development Data are relevant to likely end users. The Bank implements consultation and feedback mechanisms to (i) understand specific needs and interests for Bank Development Data, and (ii) prioritize or adapt the Bank's data activities and allocate resources accordingly." (Principle 1.e of the Policy)

The Bank is committed to ensuring that its development data are relevant to the needs of identified likely end users, including policymakers, researchers, and development practitioners. To achieve this, the Bank actively engages in consultation and feedback processes to understand user priorities, information needs, and emerging data demands.

2.5.2 Instructions

This principle applies mainly to **data products** but may also be applied to **data processes** and **systems and tools**.

The compliance report should explain:

- how the product or process meets the needs of its intended or likely users,
- how user engagement, consultation, and feedback are incorporated, and
- how relevance is reviewed and maintained over time.

Provide concrete examples, evidence, and references to consultations, user analytics, design decisions, and versioned documentation.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Identification of Users and Needs**

Demonstrate that intended or likely users are clearly understood and their needs inform the product or process:

- Identify and describe the primary and secondary user groups of the data product or process. Explain how the data addresses their needs; if no specific group was targeted, describe the anticipated audience and uses.
- Provide evidence of how user information needs and priorities were documented.
- Indicate whether gap analyses were conducted to compare user needs with existing offerings, and describe the findings.
- Provide evidence of uptake or use by the intended audiences.
- Report on any assessment of emerging trends or topics relevant to target audiences.

- Indicate whether different user skill levels and capacities were considered, and whether user profiles informed design.
- Explain the process used to decide on changes to content or scope, and describe how users were informed of these changes.

2. **Consultation and Engagement**

Show that user engagement informs planning and development:

- State whether regular engagement with users took place (e.g., surveys, interviews, workshops, advisory groups), and provide supporting evidence.
- Describe mechanisms used to gather feedback during development stages.
- Explain how users and stakeholders were consulted in defining objectives, scope, and format.
- Confirm whether consultation processes and results were documented transparently.
  Indicate whether diverse perspectives, including marginalized or underrepresented groups, were included.
- Describe how consultation outcomes were integrated into planning.

3. **Alignment with Policy and Development Priorities**

Ensure the product or process supports strategic priorities and development needs:

- Justify the creation of the product or process, citing the source (user request, policy mandate, internal initiative).
- Map the data content and coverage to relevant policy frameworks or development goals.
- Describe any linkage to a Bank or unit mandate.
- Indicate whether periodic reviews are conducted to ensure alignment with organizational and sector priorities, and provide evidence.
- Describe how the product or process adapts to changing policy agendas or crises.

4. **Product and Process Design for Usability**

Ensure design enables intended audiences to access and use the data effectively:

- Describe design features intended to facilitate access and use by target audiences.
- Confirm whether data were provided in formats, interfaces, and with metadata that match user needs; indicate whether stakeholders defined requirements for format, accessibility, or delivery.
- Explain how language, accessibility, and cultural context considerations were addressed.
- Report whether explanatory materials were provided to help users interpret and apply the data correctly.
- Indicate whether prototypes or outputs were tested with representative users before release, and summarize findings.

5. **Feedback Mechanisms and Responsiveness**

Show that user feedback is valued, monitored, and acted upon:

- Confirm whether clear channels for user feedback were established after release.
- Indicate whether user satisfaction was assessed, and describe how findings were used for improvement.

- Describe procedures for reviewing and responding to feedback promptly.
- Report on recurring feedback themes and explain how these informed updates.
- Document actions taken in response to user input.
- Confirm whether major changes resulting from feedback were communicated publicly.

6. **Monitoring and Continuous Improvement**

Demonstrate ongoing assessment and adaptation to maintain relevance:

- Describe how the relevance of the data product or process was evaluated (e.g., usage statistics, citations, policy uptake).
- Explain how relevance assessments were integrated into broader quality reviews.
- Describe how priorities were adjusted to maintain alignment with user needs over time.
- Indicate mechanisms used to track usage (web analytics, API monitoring, citation tracking, DOI assignment).
- Report whether usage statistics were analyzed and acted upon, providing examples.
- Confirm whether relevance information (e.g., target audience, intended use) was included in the standard-compliant metadata.

2.6 Coherence and Comparability

2.6.1 Principle overview

"Bank Development Data shall aim to incorporate standard concepts, definitions, classifications, and methods that enable (i) coherence, and (ii) comparability over time and between geographies." (Principle 1.f of the Policy)

Bank development data aims to be coherent and comparable across time, sectors, and geographies to support meaningful analysis and informed decision-making. Coherence ensures consistency within and across datasets, while comparability enables users to track trends, benchmark progress, and draw valid comparisons across countries and regions. To this end, the Bank prioritizes the use of standard concepts, definitions, classifications, and methodologies that align with international best practices.

2.6.2 Instructions

This principle applies to **data products** and **data processes**.

This section should explain how the product or process ensures consistency within and across datasets, and how it enables valid comparisons over time, across sectors, and between geographies. The compliance report should describe the measures taken to use standard concepts, definitions, classifications, and methodologies—preferably aligned with international best practices—and how any changes that affect comparability are managed and communicated.

Provide concrete examples, evidence, and references to standards, metadata records, coherence checks, methodological notes, and version histories.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Use of Standard Concepts and Definitions**

Demonstrate alignment with recognized statistical concepts and consistent application across datasets:

- Report on the adoption of internationally recognized statistical concepts and definitions, where applicable.
- Indicate whether documentation of definitions is available, and provide references to the relevant standards.
- Confirm that these standards are referenced in documentation and metadata.
- Confirm whether key concepts were applied consistently across datasets and over time.
- Describe procedures for reviewing and updating definitions as standards evolve.
- Identify and explain any deviations from international standards.
- Indicate whether cross-validation processes were applied to ensure internal coherence and alignment with related datasets.

- Specify whether validations were built into the design of the process or performed ex post, and clarify whether they were automated, manual, or both, and replicable.

2. **Standard Classifications and Methodologies**

Ensure consistent classification and methodological alignment to enable comparison:

- Indicate whether internationally or nationally recognized classification systems were used, and name them; if not, explain why.
- Identify and label clearly any non-standard classifications used.
- Provide evidence that classification codes and categories were applied consistently.
- Describe procedures for adapting classifications while maintaining comparability.
- Confirm whether methodologies were documented and aligned with international best practices.

3. **Internal Coherence**

Show that related datasets are conceptually and numerically consistent:

- Report whether cross-checks were conducted to ensure consistency between related datasets.
- Confirm whether indicators and variables were aligned across different products.
- Describe procedures applied to reconcile data from multiple sources within the same domain.
- Indicate whether version control was applied to ensure consistent use of updated methodologies.
- Provide documentation of internal coherence checks and their results.

4. **Temporal Comparability**

Demonstrate that data remain comparable over time or that changes are documented and communicated:

- Confirm whether consistent data collection and processing methods were applied over time.
- Document methodological or definitional changes that affected time series.
- Indicate whether guidance, notes, or warnings were provided to users on limitations to time-series comparability.
- Describe how comparability over time was maintained when methods changed.
- Report whether data were adjusted or backcast to maintain comparability.
- Confirm that datasets are annotated with information on breaks in series.
- Confirm whether procedures were in place to ensure that trend analysis was based on comparable data.

5. **Cross-Sectoral Comparability**

Ensure data can be validly compared or integrated across sectors:

- Indicate whether harmonized frameworks were applied for multi-sector data.
- Report whether sector-specific indicators were aligned with overarching definitions.
- Describe procedures used to integrate data from different sectors without loss of meaning.
- Document differences in scope, units, or measurement methods.

- Indicate whether coordination with sectoral experts was undertaken to ensure validity of cross-sector comparisons.

6. **Geographic Comparability**

Ensure spatial harmonization and clear communication of differences:

- Confirm whether consistent geographic definitions and boundaries were applied.
- Indicate whether the Bank, national, or international geographic coding systems were used.
- Document boundary changes or redefinitions that affected data.
- Describe how comparability across geographies was maintained.
- Report on limitations in comparing geographic units.

7. **Change Management and Communication**

Show that impacts on coherence and comparability are assessed and shared with users:

- Describe procedures used to assess the impact of methodological or definitional changes on comparability.
- Document and report how such changes were communicated to users in a timely manner.
- Indicate whether historical data were made publicly available in both original and revised forms, where possible.
- Confirm whether metadata explained differences between versions or datasets.
- Provide evidence that information on coherence and comparability was included in metadata.

2.7 Statistical and Data Management Standards

2.7.1 Principle overview

"a. The Bank shall for purposes of the production of Bank Development Data apply statistical methods that (a) comply with Statistical and Data Management Standards (to the extent applicable), or (b) otherwise ensure compliance with this Policy.

b. The Bank shall continuously improve Statistical and Data Management Standards applied by the Bank. For such purpose, the Bank collaborates with international, regional and national statistical organizations, and the academia." (Principle 1.g)

The Bank applies rigorous statistical methods and data management practices that comply with established statistical and data management standards or otherwise meet the requirements of its Data Quality Policy. The Bank is committed to the continuous improvement of these standards, recognizing that data quality dep ends on evolving methodologies and technologies. To this end, the Bank actively collaborates with international, regional, and national statistical organizations, as well as academic institutions, to align with global best practices and contribute to the advancement of the statistical and data science communities.

2.7.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

The compliance report should demonstrate:

- how established statistical and data management standards are applied.
- how methodological choices align with recognized best practices.
- how standards and practices evolve in response to new technologies and knowledge, and
- how collaboration with external expert communities supports improvement.

Provide concrete evidence such as methodological documentation, peer reviews, version-controlled metadata, change logs, and references to standards or guidelines adopted.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Compliance with Established Standards**

Demonstrate adherence to recognized statistical and data management standards, or justify deviations when necessary:

- Confirm whether the data process or product comply with recognized international or national statistical standards.
- Where international standards existed but have not been adopted, provide clear justification for any deviations.

- Indicate whether established data management standards are applied for storage, metadata, and interoperability.
- Confirm whether adopted standards are documented in the metadata.
- Describe procedures used to ensure consistent application of standards across all stages of the data life cycle.

2. **Statistical Methods**

Show that statistical methods are scientifically sound, well-documented, reviewed, and appropriate for the intended purpose:

- Confirm whether appropriate, scientifically sound statistical techniques were used for data collection, processing, and analysis.
- Indicate whether methods are suited to the purpose, data type, and user needs.
- Describe the process for peer or expert review of methods prior to implementation.
- Explain how contested methods are addressed.
- Confirm whether methodological choices—including rationale and references—are documented and made available to users.
- Indicate whether procedures are in place for updating methods as new standards or best practices emerged.
- For sample surveys: describe measures used to ensure accuracy, reliability, and representativeness, and indicate whether sample calibration methods are applied. Provide summary information on response rates by stratum, indicating whether non-response resulted in issued of accuracy or potential biases.
- For modeled estimates: explain how accuracy is validated, and indicate whether Testing, Evaluation, Validation, and Verification (TEVV) or equivalent procedures are applied.
- For geographic datasets and cartographic products: describe how positional and thematic accuracy is assessed, and confirm whether published cartographic products are endorsed by the Bank's Cartography unit.
- Confirm whether all assessments of statistical methods are documented and shared with users.
- Provide references to methodological papers, guidelines, or best-practice standards where relevant.
- Confirm whether information on standards and methods is included in the standard-compliant metadata.

3. **Data Management Practices**

Ensure data and metadata are governed through standard-compliant, traceable, and integrity-preserving practices:

- Confirm whether standard-compliant structural metadata (data structure definitions for indicators, and variable description in DDI Codebook for microdata) is available to describe datasets.
- Indicate whether version control was implemented for datasets and related documentation, specifying whether semantic versioning was used.
- Outline procedures for ensuring data integrity and minimizing the risk of data loss.
- Confirm whether clear rules were applied for naming, storing, and archiving datasets.
- Explain how potential outliers or anomalies in time series and input data were detected (e.g., rule-based checks, anomaly detection algorithms).
- Indicate what actions were taken to address anomalies—whether they were corrected, adjusted, or explained.

- Confirm whether data annotations were included in the dataset to explain anomalies, specifying from which version onward they applied.
- Indicate whether corrections to data were logged in a transparent change history.

4. **Continuous Improvement**

Demonstrate active monitoring of emerging practices, tools, and technologies to enhance data quality:

- Confirm whether the project team regularly monitors developments in statistical and data management methodologies.
- Describe processes for evaluating and integrating new technologies or tools.
- Explain the role of AI and automation in the production process.
- Indicate whether feedback mechanisms are in place to assess the effectiveness of improvements.
- Confirm whether feedback on detected quality issues is provided to third party data providers.

5. **Collaboration and External Engagement**

Show how external partnerships contribute to methodological alignment and leadership in global data standards:

- Describe collaborations with academic institutions and research bodies on methodological innovation.
- Confirm whether experiences and best practices are shared with peer organizations.
- Indicate whether relevant recommendations from external expert bodies were adopted.
- Describe contributions made to the development of global standards and guidance.

2.8 Access, Dissemination and Storage

2.8.1 Principle overview

"a. The Bank strives to maximize public and timely access to, and use of Bank Development Data on an equitable and non-discriminatory basis.

b. To the extent reasonably possible and appropriate taking into account potential ethical, legal and confidentiality considerations, Bank Development Data shall be made available under an open data license.

c. Acknowledging the various needs and preferences of a wide and diverse community of potential users of Bank Development Data, the Bank adopts multiple modes of data and metadata dissemination (including, without limitation, API's).

d. Bank Development Data disseminated by the Bank shall be made available together with their related Metadata.

e. The Bank shall maintain a central, searchable and publicly accessible data catalogue for Bank Development Data and ensure that Staff (i) document Data Sets, and (ii) deposit related Metadata accordingly.

f. Any dissemination by the Bank of Bank Development Data shall comply with applicable Bank policies, including the Access to Information Policy." (Principle 1.h of the Policy)

The Bank is committed to maximizing equitable, timely, and public access to its development data, making them available on a non-discriminatory basis. Wherever appropriate and feasible—while respecting ethical, legal, and confidentiality constraints—data are released under open data licenses to encourage broad use and reuse. Recognizing the diverse needs of users, the Bank disseminates data and metadata through multiple formats and channels, including APIs and user-friendly interfaces. All disseminated data are accompanied by relevant metadata to ensure transparency and usability. All dissemination activities align with applicable Bank policies, including the Policy on Access to Information and related procedures and guidance.

2.8.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

The compliance report should demonstrate:

- how access to data is equitable, timely, and aligned with open data standards,
- how dissemination channels, formats, and interfaces support diverse users,
- how metadata ensures transparency and usability, and
- how privacy, security, licensing, and other obligations are respected.

Provide concrete evidence such as URLs, screenshots, usage reports, versioned release history, accessibility testing results, and licensing statements.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Equitable and Non-Discriminatory Access**

Ensure all users have equal opportunity to obtain data, without preferential treatment unless justified.

- Confirm whether all users have equal access to data at the same time.
- Indicate whether preferential access is granted to specific groups, and if so, provide the legal or policy justification.
- Confirm whether dissemination complies with the Bank's Policy on Access to Information.
- Describe accessibility measures provided for users with disabilities (e.g., compliance with WCAG or ADA).
- Report on how access requests and fulfillment are monitored and documented.
- Confirm whether licensing terms are open, clearly communicated, and documented (e.g., reuse, repurposing, commercial use).

2. **Timely Release of Data**

Ensure release practices are prompt, predictable, and responsive to user needs:

- Indicate whether clear deadlines and priorities guide data release.
- Describe workflows used to ensure timely release, noting whether automation is applied.
- Confirm whether preliminary results, if released, are handled responsibly.
- Report whether embargoes were applied and explain their duration and justification.
- Confirm whether release schedules are published, adhered to, and made available online in machine-readable format.
- Indicate whether data are disseminated promptly after validation.
- Document and explain any delays in release.
- Describe whether urgent public needs trigger expedited release, and confirm that equal-access rules are not breached.
- Indicate whether timely updates are issued when data were revised or corrected.

3. **Open Data and Licensing**

Apply open licensing wherever feasible and communicate any restrictions transparently:

- Confirm whether open data licenses (e.g., CC BY) or open source licensing for applications and tools are applied wherever legally and ethically feasible, after consultation with ITS and LEGIA for applications and tools.
- Report any restrictions applied to access, with their justifications.
- Indicate whether licensing terms are clearly displayed.

4. **Dissemination Channels and Formats**

Ensure data are discoverable, accessible, reusable, and delivered through effective channels and tools:

- Describe search tools provided for data discovery, including semantic and AI-assisted functions.
- Indicate whether main dissemination channels (websites, dashboards, catalogs, APIs) are documented and linked.

- Confirm whether data are released in open formats (e.g., CSV, JSON)
- Conform that persistent unique identifiers (DOI or equivalent) are applied to all Bank datasets.
- Report whether data products are distributed through multiple channels and promoted via communication campaigns.
- Indicate whether partnerships with third-party redistributors enhance visibility.
- Describe measures applied to improve online discoverability (e.g., SEO).
- Report on whether user interfaces are designed with a user-centric approach and UX expertise.
- Confirm whether visualizations follow best practices and accessibility standards (e.g., WCAG), aligned with the Bank's style guide, and included downloadable underlying data.
- Indicate whether multilingual or localized dissemination is implemented where appropriate.
- Describe assessments of dissemination effectiveness and reach, and report on actions taken.
- Confirm whether datasets are included in main World Bank catalogs (e.g., DDH, Data360).

5. **Metadata and Supporting Information**

Publish complete, standardized metadata that fully enables correct data interpretation:

- Confirm whether every dataset is accompanied by comprehensive metadata.
- Indicate whether metadata covers concepts, definitions, sources, methodologies, and limitations.
- Confirm whether standardized schemas and controlled vocabularies are applied.
- Indicate whether metadata are updated alongside data releases.
- Report whether guidance materials were provided to help users interpret data correctly.

6. **Ethical, Legal, and Confidentiality Compliance**

Ensure only data cleared for safe public release are disseminated:

- Describe how confidentiality risks were assessed prior to release.
- Indicate whether anonymization or aggregation was applied where required.
- Confirm whether privacy laws and contractual obligations are respected.
- Indicate whether ethical clearance was obtained and documented where required.
- Report whether redistribution by third parties is monitored and supported.

7. **Storage and Backup**

Protect published datasets through secure, reliable preservation and access management:

- Describe processes used for secure, well-documented data storage, access control, and archiving.
- Confirm whether secure storage is ensured for datasets and metadata.
  Indicate whether regular backups are performed in multiple formats and locations.
- Report whether data repositories are covered by disaster recovery plans.
- Confirm whether long-term preservation procedures are implemented.

- Indicate whether version control is applied to maintain access to historical releases.

8. **Continuous Improvement in Access and Dissemination**

Evaluate and enhance user experience, accessibility, and technology use over time:

- Report whether user needs and satisfaction are reviewed regularly.
- Indicate whether dataset usage metrics are analyzed and used to inform improvements.
- Describe whether user groups are consulted on accessibility and formats.
- Confirm whether new dissemination technologies are adopted where appropriate.

2.9 Efficient Data Collection

2.9.1 Principle overview

"Any decision by the Bank to collect data or have data collected on the Bank's behalf (including, without limitation, a decision on the selection of data sources and the design of data collection methods and instruments) shall take into account, among other things, (a) cost-benefit, (b) the potential burden on respondents, and (c) subject to article 1(j), section III hereof, availability of suitable Third Party Data." (Principle 1.i of the Policy)

The Bank ensures that all decisions to collect data—whether directly or through third parties—are guided by principles of efficiency and responsibility. This includes carefully weighing the costs and benefits of data collection activities, minimizing the burden on respondents, and considering the availability and suitability of existing third-party data before initiating new efforts. The design of data sources, methods, and instruments is informed by these considerations to avoid duplication, optimize resource use, and maintain the highest possible quality.

2.9.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

This section of the compliance report should be completed **only when data are collected or acquired**—either directly (e.g., surveys conducted or commissioned by the Bank) or indirectly (e.g., requesting data from providers, purchasing datasets, accessing APIs, or web scraping).

The report should demonstrate that data collection was:

- necessary for the intended purpose,
- cost-effective and proportionate,
- minimally burdensome to respondents,
- coordinated to avoid duplicating existing efforts, and
- designed to achieve required quality efficiently.

Programs must describe what was done, provide supporting evidence, and explain any limitations or deviations.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Needs Assessment Before Collection**

Show that data needs were clearly defined and existing alternatives were thoroughly evaluated before collecting or acquiring new data:

- Identify and describe the main sources of data used.

- Document the purpose and expected uses of the data.
- Indicate whether stakeholder consultations were held to confirm necessity and scope, and summarize their outcomes.
- Indicate whether existing sources were assessed for adequacy, and justify any decision to conduct new collection.
- Report on the quality, coverage, and timeliness of potential third-party sources considered.
- If new data were collected, explain why existing alternatives were insufficient.
- Document decision-making processes that justified the new collection.

2. **Cost–Benefit Analysis**

Demonstrate that expected benefits justified the resources and burden associated with data collection:

- Estimate and report financial, human, and time costs of the data collection.
- Indicate expected benefits and potential impacts.
- Compare costs and benefits across alternative approaches, and describe the rationale for the chosen option.
- Document trade-offs made in scope or methodology.
- Report on whether benefits achieved to date align with initial expectations.

3. **Minimizing Respondent Burden**

Show that burden on data providers was measured, monitored, and reduced wherever possible:

- Specify how each data source was obtained (e.g., surveys, purchase, download, API, scraping).
- Describe measures taken to reduce burden on respondents or providers, such as coordinating and scheduling requests, limiting frequency of collection and justifying necessity, and streamlining acquisition processes and applying technology efficiently.
- Indicate whether burden assessments (e.g., paradata analysis) were conducted and acted upon.
- Estimate the time and effort required from respondents.
- Describe coordination with other organizations to avoid duplicate requests.
- Indicate whether sampling, administrative data, or other methods were used to reduce direct burden.
- Report on how questionnaires and submission processes were streamlined.
- Describe the instructions and support provided to respondents.

4. **Use of Existing Data**

Demonstrate strong preference toward leveraging high-quality third-party data before collecting new data:

- Report whether third-party data sources were identified and evaluated.
- Indicate whether quality, licensing, and suitability were verified.
- Describe contractual arrangements for third-party data and contingency plans in case provision ends.
- Assess and report on comparability of third-party data with Bank standards and needs.

- Indicate whether third-party data were integrated into Bank systems where feasible.

5. **Design of Data Sources, Methods, and Instruments**

Show that collection methods ensure quality while optimizing efficiency and feasibility:

- Describe how methodologies were designed to meet quality requirements at the lowest feasible cost.
- Indicate whether instruments (e.g., pilot surveys) were tested before full rollout.
- Report on use of technology to improve efficiency (e.g., mobile collection, automated scraping).
- Indicate how methods were adapted to local contexts while ensuring comparability.

6. **Avoiding Duplication**

Show that coordination occurred to prevent redundant efforts and maximize shared value:

- Report whether existing and planned data collections were reviewed to identify overlaps.
- Describe coordination with other Bank units and external partners.
- Document the unique contribution of the new collection.
- Indicate whether procedures were established to combine efforts where feasible.
- Report on monitoring measures to ensure duplication does not emerge over time.

7. **Quality Assurance in Collection**

Show that quality controls were applied during acquisition, not only post-processing:

- Describe how data quality was monitored during collection or acquisition.
- Indicate whether completeness, accuracy, and consistency were checked at the collection stage.
- Report on training provided to data collectors and staff handling third-party sources.
- Describe feedback loops established to adjust data collection when issues were detected.

2.10 Assessment of Third-Party Data

2.10.1 Principle overview

"In cases where the Bank contemplates the dissemination by the Bank of Third Party Data, or the use of Third Party Data as an input to the production of Bank Development Data, the applicable Unit shall document its assessment of the suitability of such Third Party Data for the intended purpose, addressing, among other things, the following:

a. whether quality of the Third Party Data is sufficient taking into account the principles that apply to Bank Development Data as per this Policy;

b. whether use of the Third Party Data is appropriate from a legal, ethical and technical perspective." (Principle 1.c of the Policy)

When the Bank considers using or disseminating third-party data—whether as inputs to bank development data or as stand-alone products—it conducts a thorough assessment to ensure their suitability for the intended purpose. This includes evaluating whether the quality of the data aligns with the principles set forth in the Bank's Data Quality Policy, such as accuracy, relevance, and transparency. The assessment also addresses legal, ethical, and technical considerations to ensure responsible use and compliance with applicable standards. The evaluation process must be documented to promote accountability and maintain the integrity and credibility of the Bank's data outputs.

2.10.2 Instructions

This principle applies to **data products** and **data processes**.

This section of the compliance report should describe:

- the intended purpose and scope of external data use,
- the quality and methodological rigor of the source,
- legal, ethical, and technical compliance, and
- the documented decision-making and approval processes followed.

Evidence should be provided to support assessments and justify continued reliance on third-party sources.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Purpose and Scope**

Show that the intended use and benefits are clearly defined and justified:

- Define the intended use of the third-party data (e.g., as input to a Bank product or as a stand-alone product).
- List the sources used and describe the criteria applied for their selection.

- Document the scope, coverage, expected benefits, and degree of dependency on each provider.
- Assess whether the data met identified user needs, and compare against possible alternatives.

2. **Quality Assessment**

Demonstrate that the data meet required quality standards for the intended use:

- Evaluate and report on the independence, impartiality, and robustness of the provider's methodology.
- Assess accuracy, timeliness, completeness, and internal consistency of the data.
- Review and document comparability with other relevant datasets.
- Verify and report on the transparency of methodology and source documentation.
- Assess the dataset against relevant principles of the Bank's Data Quality Policy (e.g., relevance, impartiality, verifiability).
- Document any limitations, uncertainties, or caveats identified.

3. **Legal and Licensing Review**

Show that the Bank has the right to use, transform, and disseminate the data as required:

- Verify usage rights, licensing terms, and redistribution permissions, and indicate any constraints.
- Confirm compliance with copyright, intellectual property, and contractual obligations.
- Document restrictions clearly in dissemination materials.
- Report on procedures for renewing or renegotiating licenses where applicable.

4. **Ethical and Confidentiality Considerations**

Ensure that external data were collected and handled in accordance with ethical standards:

- Review the provider's data collection methods, including consent processes where relevant.
- Confirm compliance with applicable laws, ethical standards, and Bank policies.
- Assess risks to privacy, confidentiality, or vulnerable groups.
- Indicate whether anonymization or aggregation was applied where necessary.
- Document ethical clearance when required.

5. **Technical Compatibility**

Show that the data can be securely integrated and managed within Bank systems:

- Verify and report on the formats, metadata completeness, and interoperability with Bank systems.
- Confirm compatibility with Bank storage, processing, and dissemination platforms.
- Review and document the provider's data security provisions.
- Establish and describe procedures for technical transformations required to integrate the data.

6. **Provider Assessment**

Demonstrate that the provider is credible and capable of consistent delivery:

- Review and report on the provider's reputation, expertise, and track record.
- Evaluate the provider's quality assurance processes and transparency.
- Assess alignment with international best practices.
- Indicate how provider performance is monitored over time.

7. **Documentation and Accountability**

Show that the evaluation process is transparently documented and monitored:

- Confirm that external data sources are clearly identified in public documentation.
- Document the assessment process, findings, and any reasons for rejecting sources.
- Confirm that records of contracts, correspondence, and review reports are maintained.
- Establish and report on approval procedures applied before adoption or dissemination.
- Indicate whether assessment outcomes were publicly disclosed, where appropriate.
- Report on periodic reassessments of ongoing third-party data sources.

2.11 Metadata Standards

2.11.1 Principle overview

"The Bank's processes and systems used for purposes of Data Curation and dissemination of Development Data shall comply with the Metadata Standards." (Principle 1.k of the Policy)

The Bank ensures that all processes and systems used for data curation and dissemination comply with established metadata standards. These standards enable clear, consistent, and comprehensive documentation of data, including information on definitions, methodologies, classifications, sources, and data limitations. Adherence to metadata standards enhances transparency, facilitates data discovery and interpretation, and supports interoperability across systems.

2.11.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

This section of the compliance report should describe:

- which metadata standards were applied,
- how metadata were created, maintained, and validated,
- how they are integrated into systems and workflows, and
- how users can access and understand metadata.

Where the Bank redistributes third-party data, metadata must be aligned to Bank standards to support consistent documentation and usability.

Evidence and examples should be provided where applicable.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Adoption of Metadata Standards**

Demonstrate that metadata standards were identified, applied, and justified:

- Identify the relevant World Bank metadata standards applied to the dataset.
- Describe how metadata were generated and by whom.
- Justify and document any deviations from the Bank's metadata standards.

2. **Comprehensive Metadata Documentation**

Show that metadata fully inform users about how to understand and use the data responsibly:

- Confirm whether definitions, methodologies, classifications, sources, and other essential elements are included in the metadata.
- Confirm whether data limitations, assumptions, and quality issues are clearly documented.
- Indicate whether version control is applied to metadata in parallel with datasets.
- Indicate whether metadata are updated whenever datasets are revised or corrected.

3. **System and Process Integration**

Demonstrate that metadata are captured consistently and managed through controlled processes:

- Describe how metadata are captured (e.g., automatically, systematically, or manually) within data curation systems.
- Indicate whether structured, machine-readable formats (e.g., JSON) are used to support discovery and retrieval.
- Explain how metadata processes are aligned with data processing and dissemination workflows.
- Document procedures used to verify the completeness and accuracy of metadata entries.

4. **Accessibility and Usability of Metadata**

Show that metadata are easy to find, understand, and use—both by humans and machines:

- Confirm that metadata are made available to users alongside datasets.
- Confirm that metadata are provided in both human-readable and machine-readable formats.
- Indicate the access channels through which metadata are available (e.g., APIs, downloads, catalogs).
- Confirm whether metadata are included in bulk download packages where applicable.
- Describe how metadata are presented in user-friendly formats for both technical and non-technical audiences.

2.12 Inter-operability

2.12.1 Principle overview

"a. The Bank aims to achieve maximum inter-operability of its data systems. For such purpose the Bank (i) applies an 'API-first' approach (meaning that all of a platform's functionality is accessible through APIs and available for other systems to leverage), and (ii) otherwise ensures that its data systems are built making use of common tools.

b. The Bank ensures that its data and subject-matter specialists and IT experts closely collaborate in the development of data management and dissemination platforms." (Principle 1.I of the Policy)

The Bank is committed to ensuring that its data systems are interoperable, enabling seamless integration, exchange, and reuse of data across platforms and institutions. To support this, the Bank adopts an "API-first" approach, designing its systems so that all core functionalities are accessible through standardized APIs, allowing other systems to interact with them efficiently.

2.12.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

This section of the compliance report should describe:

- how systems are designed to ensure interoperability across internal and external platforms,
- how APIs are implemented and documented,
- how consistent data and metadata standards support exchange and reuse, and
- how automated and secure data exchange is enabled and monitored.

Evidence, examples, and system documentation references should be included where relevant.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **API Availability and Management**

Demonstrate that APIs enable direct, secure, and reliable access to data and metadata:

- Indicate whether the data product is accessible through a platform that provides both data and metadata via an API.
- Specify the API type (e.g., RESTful API) and describe its key features.
- Confirm that public or internal API documentation is available, including endpoints and data models.
- Confirm that version control and backward compatibility for APIs are ensured.

- Describe the authentication, authorization, and security measures implemented.
- Indicate whether API usage and performance are monitored.

2. **API Documentation and Access**

Show that APIs are understandable, discoverable, and usable by intended audiences:

- Provide a link to the API documentation.
- Describe any restrictions on API access (e.g., authentication, quotas, licensing).
- Confirm whether all core functionalities are available through the API for both external users and system administrators, including push and pull operations, dissemination, and maintenance.

3. **System Interoperability**

Describe architecture and design practices that ensure systems can integrate effectively with others:

- Confirm that core functionalities are designed to be accessible via standardized APIs.
- Describe the use of widely accepted data exchange protocols and formats (e.g., JSON, XML, SDMX).
- Indicate whether a modular system architecture has been implemented to facilitate integration with other systems.
- Describe procedures for ensuring that new systems or updates maintain interoperability.
- List the external systems and platforms with which the Bank's systems interoperate, and describe the nature of these integrations (e.g., automated data exchange, real-time synchronization).

4. **Data Standards and Harmonization**

Demonstrate consistency of data semantics and structures that enable cross-system understanding:

- Confirm that consistent data standards are applied across platforms.
- Indicate whether controlled vocabularies, code lists, and classifications are used to enable cross-system understanding.
- Confirm alignment with metadata standards to support integration.
- Describe how differences in data structures or semantics between systems are documented.
- Explain the procedures used to reconcile or map heterogeneous datasets.

5. **Data Exchange and Reuse**

Show that systems are designed for automation-friendly data sharing and retrieval:

- Describe the mechanisms in place to allow automated data exchange between internal and external systems.
- Confirm whether standard import and export procedures are supported.
- Document integration workflows with partner or client systems.
- Describe how interoperability issues are tracked and resolved as they arise.

2.13 Responsible Innovation

2.13.1 Principle overview

"a. Recognizing the importance of responsible innovation, the Bank shall ensure that the adoption by the Bank of new technologies (including, without limitation, Artificial Intelligence) in the Data Management Lifecycle shall comply with the Data Quality Framework.

b. The Bank shall ensure that subject matter specialists (including statisticians and data scientists) are closely involved in the design, deployment and evaluation of such technologies as referred to in paragraph (a) of this article 1(m)." (Principle 1.m of the Policy)

The Bank embraces innovation in its data practices while ensuring that the adoption of new technologies—including artificial intelligence and other advanced tools—aligns with its Development Data Quality Policy. Innovation is pursued responsibly, with careful consideration of its impact on data quality, integrity, and ethical standards throughout the data life cycle. To ensure that technological advancements serve their intended purpose and uphold professional standards, the Bank involves subject-matter experts, including statisticians and data scientists, in the design, implementation, and evaluation of these tools.

2.13.2 Instructions

This principle applies to **data products, data processes,** and **systems and tools**.

This section of the compliance report should explain how innovative or emerging technologies are introduced, validated, governed, and monitored to:

- enhance efficiency and insight,
- maintain accuracy, transparency, and trustworthiness,
- comply with ethical, legal, and professional standards.

It should reference approvals, risk assessments, and governance documentation where applicable.

The dimensions listed below may be considered for assessment; the compliance report is not required to adhere to this particular structure.

1. **Identification and Use of Innovative Practices**

Describe what technologies are used, how they are evaluated, and how they affect data quality:

- Indicate whether generative AI or other emerging technologies are used at any stage of the data life cycle.
- Specify the technology applied, the implementing party, and the evaluation process followed.
- Describe how new technology is used and how outputs are validated.

- Assess potential impacts on data accuracy, reliability, relevance, and transparency.
- Document alignment with the Bank's policies, including approvals and periodic compliance reviews.

2. **Ethical Standards and Risk Mitigation**

Show that responsible innovation safeguards privacy, fairness, and trust:

- Evaluate ethical considerations prior to adopting new methods and tools.
- Assess risks related to privacy, bias, and fairness.
- Document the mitigation measures applied to address identified risks.
- Ensure procedures are established to maintain ethical standards throughout implementation.
- Conduct periodic audits or reviews of ethical performance.

3. **Expert Involvement and Oversight**

Ensure qualified professionals guide innovation decisions and outcomes:

- Involve statisticians, data scientists, and domain experts in the design, testing, and implementation of innovative technologies (AI or other).
- Document expert review and recommendations.
- Establish procedures for ongoing expert oversight during pilot phases and full deployment.
- Report on collaboration with internal or external advisory groups to provide specialized expertise.

4. **Evaluation, Monitoring, and Governance**

Demonstrate that innovation is governed by robust controls and performance checks:

- Describe testing and validation of tools before their full implementation.
- Describe how performance against intended objectives and impacts on data quality and integrity is monitored.
- Document lessons learned and adjustments made during implementation.
- Identify technological, operational, and ethical risks and document mitigation measures.
- Ensure governance frameworks are in place to approve, monitor, and review innovation projects.
- Define clear roles and responsibilities for decision-making and oversight.
- Report innovation outcomes to management and relevant stakeholders.

5. **Continuous Improvement and Organizational Learning**

Show how the Bank learns from innovation and enhances practices over time:

- Explain how user feedback and performance evaluations are exploited.
- Describe how methods and tools are updated in line with best practices and technological developments.
- Describe training provided to staff on the responsible use of new technologies.
- Document improvements, innovations, and lessons learned to support organizational learning.

3. Summary and Actions Taken or Planned

A compliance report is not meant to present a picture of perfection. It should provide evidence of compliance with core quality principles but also reflect a commitment to continuous improvement by identifying both the steps already taken and the actions planned to enhance the quality of the data product, process, or system.

Based on the self-assessment conducted for this report, provide a summary of the self-assessed status of compliance, and clearly outline:

- Actions already taken to address identified quality issues or strengthen good practices
- Actions planned or underway, with details on what is certain (e.g., already approved or resourced) and what is still under consideration

When possible, indicate:

- The expected timeline for planned actions
- The responsible parties or teams
- Any dependencies or risks that could affect implementation

4. Review by the Office of the Chief Statistician

This section should document the review process carried out by the Office of the WBG Chief Statistician. It should specify who conducted the review, the date of the review, and the scope of what was assessed. The section should summarize the main outcomes of the review, including confirmation that the submitted report addressed the feedback received during the drafting process, and should explicitly state that no objection was formulated by the Chief Statistician.

If the Office of the Chief Statistician issued recommendations for future improvement, these should be briefly described, along with an indication of whether they have been incorporated into the plan for further enhancements outlined in Section 3.

5. References and Links

This section should provide a list of relevant documents, tools, websites, and other resources that support or illustrate the practices described in the preceding sections. Each item should be identified with its title, source or authoring entity, and a short description (one to two sentences) summarizing its relevance. Where possible, provide direct links to the resources to facilitate access.

Distinguish between resources that are exclusively available within the Bank and those that are accessible to the public.

[1] See https://github.com/worldbank/quality-in-modeled-estimates